

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-adce21c234bd038ee.jsonl`  
- SHA-256: `4550926ec470d75f40d60846066d0f4adcbec0783db52705c0dc704c43676c2d`  
- Source modified: `2026-06-06T17:31:12+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:30:26.465Z)

Source Extractor

Research question: "Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ? especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio)."

Fetch and extract key claims from this source:

****URL:**** https://arxiv.org/html/2510.18405

****Title:**** OCR-Guided Wicket-Taking Delivery Segmentation in Cricket Video (YOLOv8 + scoreboard OCR, arXiv 2510.18405)

****Found via:**** classical/lightweight search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:30:31.607Z)

I'll fetch the source page to extract key claims relevant to the research question.

3. user (2026-06-06T17:30:49.031Z)

Analysis of Cricket Video Analysis Paper

****Publication Details:****

- ****Date:**** March 18, 2026 (arXiv:2510.18405v2)

- ****Authors:**** Joy Karmoker, Masum Billah, Mst Jannatun Ferdous, Akif Islam, Mohd Ruhul Ameen, Md. Omar Faruqe

- ****License:**** arXiv.org perpetual non-exclusive license

(1) Wicket-Taking Delivery Segmentation Method

****Does it use YOLOv8 + scoreboard OCR?****

Yes. The system employs a three-stage pipeline:

> "The pipeline consists of three sequential stages: OCR-guided wicket-event segmentation, YOLOv8-based pitch and ball detection, and trajectory-based dismissal-zone analysis."

****How OCR-guided temporal segmentation works:****

The system samples frames at 0.1-second intervals and extracts scorecard regions. Text undergoes preprocessing including "grayscale conversion, power transformation, and morphological operations" to enhance readability. A wicket event triggers when "the recognized wicket count increases relative to the preceding valid score instance."

> "Once such an increment is detected, the corresponding delivery segment is extracted and saved."

(2) Labels/Training Data Required

****Pitch Detection Dataset:****

- 951 annotated images from Roboflow

- 797 training (83.8%), 130 validation (13.7%), 12 test (1.3%)
- Single class ("Pitch")

****Ball Detection Dataset:****

- 257 annotated images from Roboflow
- 210 training (81.7%), 28 validation (10.9%), 27 test (10.5%)
- Single class ("Ball")

> "Annotations are provided in YOLO format with a single target class per dataset."

(3) Reported Accuracy Metrics

Model	Precision	Recall	mAP50	mAP50-95
Pitch Detection (Best)	0.999	1.000	0.995	0.984
Ball Detection (Transfer Learning)	0.968	0.978	0.9918	0.930
Ball Detection (From Scratch)	0.725	0.725	0.773	0.374

> "The transfer-learning model achieved 0.968 precision, 0.978 recall, and 0.9918 mAP50."

Transfer learning showed "+33.5% precision improvement" over training from scratch.

(4) Compute Cost / Runtime / Hardware

****Not disclosed.**** The paper provides no information on inference time, computational requirements, or recommended hardware specifications.

(5) License Information

- ****Models:**** Not specified in paper
- ****Code:**** No repository or code availability mentioned
- ****Datasets:**** Both pitch and ball detection datasets sourced from Roboflow (public datasets referenced but licensing not discussed)

> "The pitch detection dataset contains 951 annotated images obtained from Roboflow, while the ball detection dataset contains 257 annotated images from Roboflow."

(6) Scoreboard/Score-Overlay OCR for Event Detection

Yes, explicitly used:

> "A wicket event is declared when the recognized wicket count increases relative to the preceding valid score instance."

The scorecard region is isolated using a "predefined region of interest (ROI)...typically located near the lower portion of the frame," enabling temporal event boundary detection prior to spatial analysis.

4. assistant (2026-06-06T17:30:51.724Z)

I have the extracted content. Let me synthesize the key falsifiable claims relevant to the research question on rally/event segmentation via scoreboard OCR.

5. user (2026-06-06T17:31:01.982Z)

Structured output provided successfully

6. assistant (2026-06-06T17:31:12.901Z)

I have extracted and returned the structured output.

Summary of what I found in arXiv:2510.18405 ("OCR-Guided Wicket-Taking Delivery Segmentation in Cricket Video"):

This is a **primary research source** (peer-style arXiv preprint, v2 dated 2026-03-18, six authors). Its relevance to the rally-segmentation research question is via the **scoreboard-OCR family** of temporal segmentation, which transfers conceptually to badminton point boundaries:

- **Central finding**: score-overlay OCR can serve as the primary temporal segmentation signal ? declare an event boundary when the recognized score count increments versus the previous valid reading. This is directly analogous to detecting badminton point boundaries from a score overlay, and is fully local-runnable with no cloud dependency.
- **Lightweight, local recipe**: fixed-interval frame sampling (0.1 s), a predefined scoreboard ROI, and classical preprocessing (grayscale, power transform, morphology) before OCR ? cheap enough for the RTX 2070 Super target.
- **Low label cost**: the OCR temporal gate needs no per-event training data; only the auxiliary YOLOv8 spatial detectors used small Roboflow sets.
- **Caveats for our context** (limit how far this can be relied on): the paper discloses **no inference-time/compute numbers, no hardware, no code repo, and no model/dataset license terms** ? only the standard arXiv license. So commercial/license viability and local-compute cost are unestablished from this source, and it does not address badminton specifically.

Net: this source supports scoreboard-OCR as a complementary, cheap, fully-local point-boundary signal worth a next experiment where broadcast score overlays exist ? but it does not by itself validate accuracy or licensing for our self-hosted commercial setup.